

# VECTOR SIGNAL ANALYSIS APPLICATION

## Specifications

R&S®FSW3-KM101/FSW/FSWP/FSWT/FSMR3/FSV3/FPS/FPL1/VSE-K70 Vector Signal Analysis Application  
R&S®FSW3-KM102/FSW/FSWP/FSWT/FSMR3/FSV3/FPL1/VSE-K70M Multi-Modulation Analysis Application  
R&S®FSW3-KM103/FSW/FSWP/FSWT/FSMR3/FSV3/FPL1/VSE-K70P BER PRBS Measurement Application  
R&S®FSW3-KM104 Cross-Correlation for Vector Signal Analysis



Specifications  
Version 19.00

**ROHDE & SCHWARZ**

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# Definitions

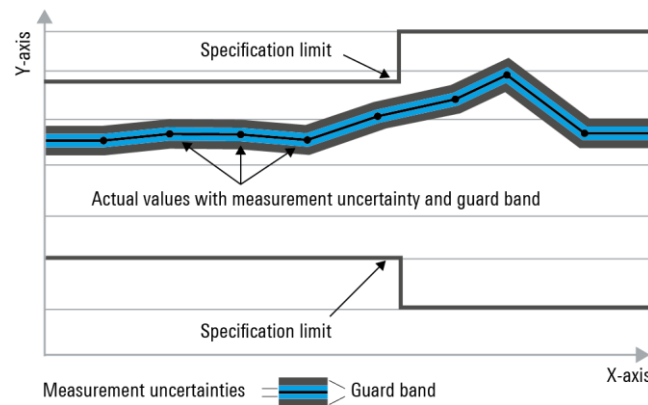
## General

Product data applies under the following conditions:

- Three hours of storage at ambient temperature followed by 30 minutes of warm-up operation
- Specified environmental conditions met
- Recommended calibration interval adhered to
- All internal automatic adjustments performed, if applicable

## Specifications with limits

Represent warranted product performance by means of a range of values for the specified parameter. These specifications are marked with limiting symbols such as  $<$ ,  $\leq$ ,  $>$ ,  $\geq$ ,  $\pm$  or descriptions such as maximum, limit of, minimum. Compliance is ensured by testing or is derived from the design. Test limits are narrowed by guard bands to take into account measurement uncertainties, drift and aging, if applicable.



## Non-traceable specifications with limits (n. trc.)

Represent product performance that is specified and tested as described under "Specifications with limits" above. However, product performance in this case cannot be warranted due to the lack of measuring equipment traceable to national metrology standards. In this case, measurements are referenced to standards used in the Rohde & Schwarz laboratories.

## Specifications without limits

Represent warranted product performance for the specified parameter. These specifications are not specially marked and represent values with no or negligible deviations from the given value (e.g. dimensions or resolution of a setting parameter). Compliance is ensured by design.

## Typical data (typ.)

Characterizes product performance by means of representative information for the given parameter. When marked with  $<$ ,  $>$  or as a range, it represents the performance met by approximately 80 % of the instruments at production time. Otherwise, it represents the mean value.

## Nominal values (nom.)

Characterize product performance by means of a representative value for the given parameter, e.g. nominal impedance. In contrast to typical data, a statistical evaluation does not take place and the parameter is not tested during production.

**Measured values (meas.)**

Characterize expected product performance by means of measurement results gained from individual samples.

**Uncertainties**

Represent limits of measurement uncertainty for a given measurand. Uncertainty is defined with a coverage factor of 2 and has been calculated in line with the rules of the Guide to the Expression of Uncertainty in Measurement (GUM), taking into account environmental conditions, aging, wear and tear.

Device settings and GUI parameters are indicated as follows: "parameter: value".

Typical data as well as nominal and measured values are not warranted by Rohde & Schwarz.

In line with the 3GPP standard, chip rates are specified in million chips per second (Mcps), whereas bit rates and symbol rates are specified in billion bit per second (Gbps), million bit per second (Mbps), thousand bit per second (kbps), million symbols per second (Msps) or thousand symbols per second (ksps), and sample rates are specified in million samples per second (Msample/s). Gbps, Mcps, Mbps, Msps, kbps, ksps and Msample/s are not SI units.

# Specifications

The specifications of the vector signal analysis application options are based on the specifications of the FSWX, FSW, R&S®FSWP, R&S®FSVA3000, R&S®FSV3000, R&S®FPS, FPL signal and spectrum analyzers, the R&S®FSWT test receiver, the R&S®FSMR3000 measuring receiver and the R&S®RTO oscilloscope. They have not been checked separately and are not verified during instrument calibration. Measurement uncertainties are given as 95 % confidence intervals. The specified level measurement errors do not take into account systematic errors due to reduced signal-to-noise ratio (SNR).

## General remarks

These specifications cover:

- R&S®FSW3-KM101/-KM102/-KM103/-KM104
- R&S®FSW-K70/-K70M/-K70P
- R&S®FSWP-K70/-K70M/-K70P
- R&S®FSWT-K70/-K70M/-K70P
- R&S®FSMR3-K70/-K70M/-K70P
- R&S®FSV3-K70/-K70M/-K70P
- R&S®FPS-K70
- R&S®FPL1-K70/-K70M/-K70P
- R&S®VSE-K70/-K70M/-K70P

The FSW, R&S®FSWP, R&S®FSWT, R&S®FSMR3, R&S®FSV3, R&S®FPS and R&S®FPL1 vector signal analysis options are summarized with the term R&S®FSx-K70.

For better readability, devices listed in the tables are abbreviated as follows: The R&S®FSWP is referred to as “FSWP”, and so on.

The R&S®FSW3 and R&S®FSx vector signal analysis options run on the device itself.

The R&S®VSE vector signal analysis options run on a PC that can be connected to the devices as specified below.

If not stated otherwise, the values in these specifications are device-specific, e.g. the same value applies to the R&S®FSW-K70 and the R&S®VSE-K70 with connected FSW.

For feature tables the following conventions apply (exemplary for -K70):

|             |                                                                                                                                                               |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| •           | feature always supported, i.e. with the R&S®VSE-K70 connected to the device and with the corresponding R&S®FSx-K70 option when running directly on the device |
| • (VSE)     | feature supported only with the R&S®VSE-K70 connected to the device; not with the corresponding R&S®FSx-K70 option when running directly on the device        |
| • (FSx-K70) | feature supported only when running directly on the device with the corresponding FSx-K70 option; not supported in the R&S®VSE-K70                            |
| –           | feature not supported with this device                                                                                                                        |

## Overview

| Measurement application                                                        | Description                                 | FSWX                                                 | FSW                         | FSWP <sup>1</sup>             | FSWT                          | FSMR3000 <sup>2</sup>          | FSVA3000/<br>FSV3000          | FPS          | FPL                           | RTO |
|--------------------------------------------------------------------------------|---------------------------------------------|------------------------------------------------------|-----------------------------|-------------------------------|-------------------------------|--------------------------------|-------------------------------|--------------|-------------------------------|-----|
| R&S®FSW3-KM101/<br>-KM102/-KM103/<br>-KM104<br><br>R&S®FSx-K70/<br>-K70M/-K70P | software that runs on device                | •<br>FSW3<br>-KM101/<br>-KM102/<br>-KM103/<br>-KM104 | •<br>FSW-K70<br>-K70M/-K70P | •<br>FSWP-K70/<br>-K70M/-K70P | •<br>FSWT-K70/<br>-K70M/-K70P | •<br>FSMR3-K70/<br>-K70M/-K70P | •<br>FSV3-K70/<br>-K70M/-K70P | •<br>FPS-K70 | •<br>FPL1-K70/<br>-K70M/-K70P | —   |
| R&S®VSE-K70/<br>-K70M/-K70P                                                    | PC software that can be connected to device | •                                                    | •                           | •                             | •                             | —                              | •                             | •            | •                             | •   |

## Vector signal analysis application

### Inputs

|                        | FSWX                                  | FSW                                   | FSWP | FSWT                         | FSMR3000 | FSVA3000/<br>FSV3000 | FPS | FPL | RTO (VSE) |
|------------------------|---------------------------------------|---------------------------------------|------|------------------------------|----------|----------------------|-----|-----|-----------|
| RF input               | •                                     | •                                     | •    | •                            | •        | •                    | •   | •   | •         |
| External frontend      | •<br>(FSW3<br>-KM101) <sup>3, 5</sup> | •<br>(FSW<br>-K70) <sup>3, 4, 5</sup> | —    | —                            | —        | • <sup>4</sup>       | —   | —   | —         |
| Digital baseband input | —                                     | •<br>(FSW-K70) <sup>6</sup>           | —    | •<br>(FSWT-K70) <sup>7</sup> | —        | —                    | —   | —   | —         |
| Analog baseband input  | —                                     | •<br>(FSW-K70) <sup>8</sup>           | —    | —                            | —        | —                    | —   | —   | —         |
| File                   | •                                     | •                                     | •    | •                            | •        | •                    | •   | •   | •         |

<sup>1</sup> R&S®FSWP-B1 option required.

<sup>2</sup> R&S®FSMR3-B1 option required.

<sup>3</sup> R&S®FE170SR supported. For details see the specifications of R&S®FE170SR (PD 3683.3694.22).

<sup>4</sup> R&S®FE44S and R&S®FE50DTR supported. For details see the specifications of R&S®FE44S (PD 3609.5545.22) and R&S®FE50DTR (PD 3609.5551.22).

<sup>5</sup> R&S®FE110SR supported. For details see the specifications of R&S®FE110SR (PD 3683.9457.22).

<sup>6</sup> Only with R&S®FSW-B17 option.

<sup>7</sup> Only with R&S®FSWT-B17 option.

<sup>8</sup> Only with R&S®FSW-B71 option.

## Signal acquisition

|                                   |                                        | FSWX                                                | FSW                                    | FSWP                      | FSWT                      | FSMR3000            | FSVA3000/<br>FSV3000                   | FPS                 | FPL                   | RTO<br>(VSE)        |
|-----------------------------------|----------------------------------------|-----------------------------------------------------|----------------------------------------|---------------------------|---------------------------|---------------------|----------------------------------------|---------------------|-----------------------|---------------------|
| Capture length                    | maximum number of symbols <sup>9</sup> | 230 000 000<br>(FSW3-KM101)                         | 230 000 000 <sup>10</sup><br>(FSW-K70) | 230 000 000<br>(FSWP-K70) | 200 000 000<br>(FSWT-K70) | 230 000 000         | 50 000 000 <sup>11</sup><br>(FSV3-K70) | 100 000 000         | 128 000<br>(FPL1-K70) | 19 000 000          |
|                                   |                                        | 100 000 000<br>(VSE)                                | 100 000 000 <sup>10</sup><br>(VSE)     | 100 000 000<br>(VSE)      | 100 000 000<br>(VSE)      |                     | 100 000 000<br>(VSE)                   |                     | 12 000 000<br>(VSE)   |                     |
| Sample rate <sup>12</sup>         | standard                               | 100 Hz to<br>5 GHz                                  | 100 Hz to<br>10 GHz                    | 100 Hz to<br>10 GHz       | 100 Hz to<br>7.2 GHz      | 100 Hz to<br>10 GHz | 100 Hz to<br>128 MHz                   | 100 Hz to<br>10 GHz | 100 Hz to<br>16 MHz   | 100 Hz to<br>10 GHz |
|                                   | up to <sup>13</sup>                    | 20 GHz                                              | 20 GHz                                 | 20 GHz                    | 7.2 GHz                   | 20 GHz              | 10 GHz                                 | 20 GHz              | 100 MHz               | 10 GHz              |
| Capture oversampling              |                                        | 2, 4, 8, 16, 32, 64, 128 <sup>14</sup>              |                                        |                           |                           |                     |                                        |                     |                       |                     |
| Symbol rate <sup>15, 16, 17</sup> | depends on capture oversampling        | sample rate/capture oversampling                    |                                        |                           |                           |                     |                                        |                     |                       |                     |
| Optimization                      |                                        | cross-correlation for EVM improvement <sup>18</sup> | —                                      |                           |                           |                     |                                        |                     |                       |                     |

<sup>9</sup> The upper limit of the capture length applies to the minimum capture oversampling. If the capture oversampling is larger, the maximum capture length is reduced proportionally, e.g. for capture oversampling = 8, it is 25 000 000 symbols for the FSW. The capture length may be restricted by the available RAM.

For additional restrictions and hardware requirements, see FSWX, FSW, R&S®FSWP, R&S®FSWT, R&S®FSMR3000, R&S®FSVA3000, R&S®FSV3000, R&S®FPS, FPL, R&S®VSE and R&S®RTO specifications.

<sup>10</sup> If the R&S®FSW-B2000 or R&S®FSW-B5000 is active, the maximum capture length is limited to 128 000 symbols.

<sup>11</sup> Firmware version 1.20 or higher required.

<sup>12</sup> Setting range.

<sup>13</sup> Depends on the hardware configuration. For details, see FSWX, FSW, R&S®FSWP, R&S®FSWT, R&S®FSMR3000, R&S®FSVA3000, R&S®FSV3000, R&S®FPS, FPL and R&S®RTO specifications.

<sup>14</sup> For large symbol rates, restricted by the maximum sample rate. Capture oversampling of 2 is only supported for PSK and QAM.

<sup>15</sup> RF input: the maximum symbol rate a measured signal is allowed to have is also limited by the usable I/Q bandwidth of the analyzer, the actual bandwidth of the measured signal (depends e.g. on filter rolloff (alpha)) and any frequency offset (FO). The following condition has to be met for an example with a raised cosine filter:  $[\text{symbol rate} \times (1 + \alpha) + 2 \times \text{FO} < \text{usable I/Q bandwidth}]$ .

<sup>16</sup> Digital baseband input with R&S®FSW-B17/R&S®FSWT-B17 option: the maximum symbol rate a measured signal is allowed to have is also limited by the digital input sample rate, the actual bandwidth of the measured signal (depends e.g. on filter rolloff (alpha)) and any frequency offset (FO).

The following condition has to be met for an example with a raised cosine filter:  $[\text{symbol rate} \times (1 + \alpha) + 2 \times \text{FO} < 0.8 \times \text{digital input sample rate/capture oversampling}]$ .

<sup>17</sup> Analog baseband input with R&S®FSW-B71 option: the maximum symbol rate a measured signal is allowed to have is also limited by the analyzer's frequency range, the actual bandwidth of the measured signal (depends e.g. on filter rolloff (alpha)) and any frequency offset (FO). The following condition has to be met for an example with a raised cosine filter:  $[0.5 \times \text{symbol rate} \times (1 + \alpha) + \text{FO} < \text{half frequency range}]$ .

<sup>18</sup> R&S®FSW3-KM104 option required. For FSWX models with two RF ports, cross-correlation is only available for RF port A.

|                         |                                                       | FSWX                                                         | FSW      | FSWP            | FSWT    | FSMR3000 | FSVA3000/<br>FSV3000   | FPS     | FPL      | RTO<br>(VSE) |
|-------------------------|-------------------------------------------------------|--------------------------------------------------------------|----------|-----------------|---------|----------|------------------------|---------|----------|--------------|
| Usable I/Q<br>bandwidth | depends on set<br>symbol rate                         | about $0.8 \times$ capture oversampling $\times$ symbol rate |          |                 |         |          |                        |         |          |              |
|                         | standard                                              | 40 MHz                                                       | 10 MHz   | 10 MHz          | 500 MHz | 10 MHz   | 28 MHz <sup>19</sup>   | 28 MHz  | 12.8 MHz | 600 MHz      |
|                         | up to <sup>13</sup>                                   | 4000 MHz                                                     | 8312 MHz | 320 MHz         | 500 MHz | 80 MHz   | 1000 MHz <sup>19</sup> | 160 MHz | 40 MHz   | 6 GHz        |
| Trigger modes           | RF<br>input                                           | free run                                                     | •        | •               | •       | •        | •                      | •       | •        | •            |
|                         |                                                       | external                                                     | •        | •               | •       | •        | •                      | •       | •        | •            |
|                         |                                                       | IF power <sup>20</sup>                                       | —        | •               | •       | •        | •                      | •       | •        | —            |
|                         |                                                       | I/Q power                                                    | •        | • <sup>21</sup> | •       | •        | •                      | —       | •        | —            |
|                         | base-<br>band<br>input                                | baseband<br>power                                            | —        | •<br>(FSW-K70)  | —       | —        | —                      | —       | —        | —            |
|                         |                                                       | burst search <sup>22</sup>                                   | •        | •               | •       | •        | •                      | •       | •        | •            |
|                         | predefined or user-<br>defined patterns <sup>23</sup> |                                                              | •        | •               | •       | •        | •                      | •       | •        | •            |

<sup>19</sup>  $f > 7.5$  GHz, with R&S®FSV3-B11 option and YIG preselector off.

<sup>20</sup> Restricted functionality at carrier frequencies  $< 50$  MHz. The IF power trigger can be also limited in bandwidth. For details, see FSW, R&S®FSWP, R&S®FSWT, R&S®FSMR3000, R&S®FSVA3000, R&S®FSV3000, R&S®FPS and FPL specifications.

<sup>21</sup> Not available for analysis bandwidth  $> 80$  MHz, if R&S®FSW-B512 is installed.

<sup>22</sup> Checks captured data for power bursts and performs analysis only on detected burst.

<sup>23</sup> Checks captured data for patterns and performs analysis only on detected pattern.

## Modulation formats

Two different PSK and QAM modulations can be measured within the same result range for R&S®FSW3-KM101, R&S®FSW-K70, R&S®FSWP-K70, R&S®FSWT-K70, R&S®FSMR3-K70, R&S®FSV3-K70, R&S®FPL1-K70 and R&S®VSE-K70.

Rotating PSK and QAM modulations, such as  $\pi/4$ -16QAM, need to have the same additional phase shift in both modulation ranges. E.g., a  $\pi/4$ -16QAM can be combined with a  $\pi/4$ -QPSK, but not with a  $3\pi/8$ -8PSK.<sup>24</sup>

| Type                                                                    | Order                                                                                                                       |
|-------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| FSK                                                                     | 2FSK                                                                                                                        |
|                                                                         | 4FSK                                                                                                                        |
|                                                                         | 8FSK                                                                                                                        |
|                                                                         | 16FSK                                                                                                                       |
|                                                                         | 32FSK                                                                                                                       |
|                                                                         | 64FSK                                                                                                                       |
| MSK                                                                     | MSK, including GMSK                                                                                                         |
|                                                                         | DMSK                                                                                                                        |
| PSK                                                                     | BPSK, $\pi/2$ -BPSK, $-\pi/2$ -BPSK, $\pi/2$ -DBPSK                                                                         |
|                                                                         | QPSK, DQPSK, $\pi/4$ -QPSK, $3\pi/4$ -QPSK, $\pi/4$ -DQPSK                                                                  |
|                                                                         | offset QPSK                                                                                                                 |
|                                                                         | shaped offset QPSK                                                                                                          |
|                                                                         | 8PSK, $3\pi/8$ -8PSK, D8PSK, $\pi/8$ -D8PSK                                                                                 |
| QAM                                                                     | 16QAM, $\pi/4$ -16QAM                                                                                                       |
|                                                                         | 32QAM, $-\pi/4$ -32QAM                                                                                                      |
|                                                                         | 64QAM                                                                                                                       |
|                                                                         | 128QAM                                                                                                                      |
|                                                                         | 256QAM                                                                                                                      |
|                                                                         | 512QAM                                                                                                                      |
|                                                                         | 1024QAM                                                                                                                     |
|                                                                         | 2048QAM <sup>25</sup>                                                                                                       |
|                                                                         | 4096QAM <sup>25</sup>                                                                                                       |
| ASK                                                                     | 2ASK                                                                                                                        |
|                                                                         | 4ASK                                                                                                                        |
| APSK                                                                    | 16APSK (DVB-S2 configurations for different code rates)                                                                     |
|                                                                         | 32APSK (DVB-S2 configurations for different code rates)                                                                     |
| User modulation (QAM, PSK)<br>(with external MAPWIZ <sup>26</sup> tool) | user-definable constellation:<br>2-ary, 4-ary, 8-ary, 16-ary, 32-ary, 64-ary, 128-ary, 256-ary<br>(static and differential) |

<sup>24</sup> Requires R&S®FSW3-KM102, R&S®FSW-K70M, R&S®FSWP-K70M, R&S®FSWT-K70M, R&S®FSMR3-K70M, R&S®FSV3-K70M, R&S®FPL1-K70M or R&S®VSE-K70M, respectively.

<sup>25</sup> Only available for FSWX, FSW, R&S®FSWP, R&S®FSWT.

<sup>26</sup> MAPWIZ is a free Rohde & Schwarz tool that can be downloaded at [www.rohde-schwarz.com](http://www.rohde-schwarz.com). It requires MATLAB.

## Predefined standards

Predefined standards can be loaded in order to preset the measurement parameters, filters and display format. Predefined standards can be changed and resaved.

|                           |                                      |                                                                                                                         |
|---------------------------|--------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| 3GPP CDMA                 | QPSK                                 | CPICH<br>(without descrambling and despreading)                                                                         |
| GSM, EDGE, EDGE Evolution | GMSK                                 | normal burst                                                                                                            |
|                           |                                      | access burst                                                                                                            |
|                           |                                      | frequency correction burst                                                                                              |
|                           |                                      | synchronization burst                                                                                                   |
|                           | 3 $\pi$ /8-8PSK                      | normal burst                                                                                                            |
|                           | 3 $\pi$ /4-QPSK                      | higher symbol rate burst with narrow and wide pulse filter                                                              |
|                           | $\pi$ /4-16QAM                       | normal burst                                                                                                            |
|                           |                                      | higher symbol rate burst with narrow and wide pulse filter                                                              |
|                           | - $\pi$ /4-32QAM                     | normal burst                                                                                                            |
|                           |                                      | higher symbol rate burst with wide pulse filter                                                                         |
| TETRA                     | $\pi$ /4-DQPSK                       | discontinuous downlink<br>continuous downlink                                                                           |
| APCO25                    | QPSK                                 | CQPSK                                                                                                                   |
|                           | 4FSK                                 | C4FM                                                                                                                    |
|                           | CPM                                  | H-CPM                                                                                                                   |
|                           | DQPSK                                | H-DQPSK                                                                                                                 |
| Bluetooth®                | 2FSK                                 | DH1                                                                                                                     |
|                           |                                      | DH3                                                                                                                     |
|                           |                                      | DH5                                                                                                                     |
|                           |                                      | low energy                                                                                                              |
|                           |                                      | 5.0 low energy                                                                                                          |
|                           |                                      | 2-DH1                                                                                                                   |
|                           | $\pi$ /4-DQPSK                       | 2-DH3                                                                                                                   |
|                           |                                      | 2-DH5                                                                                                                   |
|                           |                                      | 3-DH1                                                                                                                   |
|                           | D8PSK                                | 3-DH3                                                                                                                   |
|                           |                                      | 3-DH5                                                                                                                   |
| DOCSIS 3.0 downlink       | 64QAM                                | J83B                                                                                                                    |
|                           |                                      | Euro DOCSIS                                                                                                             |
|                           | 256QAM                               | J83B                                                                                                                    |
|                           |                                      | Euro DOCSIS                                                                                                             |
| DECT                      | 2FSK                                 | P32, fixed part<br>P32, portable part                                                                                   |
| DVB-S2                    | QPSK                                 |                                                                                                                         |
|                           | 8PSK                                 |                                                                                                                         |
|                           | 16APSK                               | only XFECFRAME                                                                                                          |
|                           | 32APSK                               | only XFECFRAME                                                                                                          |
| DVB-S2/DVB-S2X            | up to 256APSK                        | Supported via a free Rohde & Schwarz tool that preconfigures R&S®FSW3-KM101, R&S®FSx-K70 and R&S®VSE-K70, respectively. |
| DVB-RCS2 <sup>24</sup>    | $\pi$ /2-BPSK, QPSK, 8PSK, and 16QAM | Supported via a free Rohde & Schwarz tool that preconfigures R&S®FSx-K70 and R&S®VSE-K70, respectively.                 |

|                        |                    |                                                   |
|------------------------|--------------------|---------------------------------------------------|
| CDMA2000               | QPSK               | 1 × forward link <sup>27</sup>                    |
|                        | offset QPSK        | 1 × reverse link <sup>27</sup>                    |
| ZigBee (IEEE 802.15.4) | offset QPSK        | PHY for 2450 MHz band <sup>27</sup>               |
|                        | BPSK               | PHY for 915 MHz band <sup>27</sup>                |
|                        |                    | PHY for 868 MHz band <sup>27</sup>                |
| SOQPSK                 | shaped offset QPSK | shaped offset QPSK TG (IRIG 106-13) <sup>28</sup> |
| DMR                    | 4FSK               |                                                   |
| Others                 |                    | user-definable standards                          |

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<sup>27</sup> Without descrambling and dispreading.

<sup>28</sup> Without differential precoding.

## Filtering

|                                   |                               |                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------------------|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Filter types                      | transmit filter               | RC (raised cosine)<br>RRC (root raised cosine)<br>Gaussian<br>GMSK<br>linearized GMSK<br>EDGE narrow pulse shape<br>EDGE wide pulse shape<br>CDMA2000 1x forward<br>CDMA2000 1x reverse<br>APCO25 C4FM<br>APCO25 H-CPM<br>APCO25 H-DQPSK<br>APCO25 H-D8PSK narrow<br>APCO25 H-D8PSK wide<br>half sine<br>rectangular<br>shaped offset QPSK TG<br>none<br>user-definable filters designed with FILTWIZ <sup>29</sup> |
|                                   | measurement filter            | RRC<br>Gaussian<br>EDGE NSR<br>EDGE HSR (narrow pulse)<br>EDGE HSR (wide pulse)<br>rectangular<br>low ISI measurement filter <sup>30</sup><br>narrow lowpass<br>wide lowpass<br>none<br>user-definable filters designed with FILTWIZ <sup>29</sup>                                                                                                                                                                  |
|                                   | receive filter                | appropriate receive filters are automatically selected                                                                                                                                                                                                                                                                                                                                                              |
| User-selectable filter parameters |                               |                                                                                                                                                                                                                                                                                                                                                                                                                     |
| Alpha (rolloff factor)            | for RC and RRC filters        | 0.030 to 1                                                                                                                                                                                                                                                                                                                                                                                                          |
| B × T                             | for Gaussian and GMSK filters | 0.030 to 1                                                                                                                                                                                                                                                                                                                                                                                                          |

<sup>29</sup> FILTWIZ is a free Rohde & Schwarz tool that can be downloaded at [www.rohde-schwarz.com](http://www.rohde-schwarz.com). It requires MATLAB.

<sup>30</sup> Not available for shaped offset QPSK.

**Measurement parameters**

|                               |                                                                                     | FSWX    | FSW     | FSWP    | FSWT    | FSMR3000 | FSVA3000/<br>FSV3000                             | FPS     | FPL     | RTO (VSE) |
|-------------------------------|-------------------------------------------------------------------------------------|---------|---------|---------|---------|----------|--------------------------------------------------|---------|---------|-----------|
| Result length                 | maximum number of symbols <sup>31</sup>                                             | 128 000 | 128 000 | 128 000 | 128 000 | 128 000  | 128 000                                          | 128 000 | 128 000 | 128 000   |
| Coarse synchronization        |                                                                                     |         |         |         |         |          | data (based on unknown data)                     |         |         |           |
|                               | only if a synchronization pattern is found                                          |         |         |         |         |          | pattern (based on synchronization pattern)       |         |         |           |
| Fine synchronization          |                                                                                     |         |         |         |         |          | detected data (based on detected data)           |         |         |           |
|                               | only if a synchronization pattern is found                                          |         |         |         |         |          | pattern (based on synchronization pattern)       |         |         |           |
|                               | only if known data is provided<br>(see requirements for bit error rate measurement) |         |         |         |         |          | known data (based on detected transmit sequence) |         |         |           |
| EVM normalization             | only for PSK, QAM, ASK and APSK                                                     |         |         |         |         |          | mean reference power                             |         |         |           |
|                               |                                                                                     |         |         |         |         |          | maximum reference power                          |         |         |           |
|                               |                                                                                     |         |         |         |         |          | mean constellation power                         |         |         |           |
|                               |                                                                                     |         |         |         |         |          | maximum constellation power                      |         |         |           |
| Optimization                  |                                                                                     |         |         |         |         |          | minimize RMS error                               |         |         |           |
|                               |                                                                                     |         |         |         |         |          | minimize EVM                                     |         |         |           |
| Offset EVM                    | only for offset QPSK, shaped offset QPSK                                            |         |         |         |         |          | on/off                                           |         |         |           |
| Equalizer estimation          | only for PSK <sup>32</sup> , QAM, ASK, APSK and MSK                                 |         |         |         |         |          | normal                                           |         |         |           |
|                               |                                                                                     |         |         |         |         |          | tracking                                         |         |         |           |
|                               |                                                                                     |         |         |         |         |          | averaging                                        |         |         |           |
|                               |                                                                                     |         |         |         |         |          | user-defined                                     |         |         |           |
| Equalizer length              | only for PSK <sup>32</sup> , QAM, ASK, APSK and MSK                                 |         |         |         |         |          | 1 symbol to 256 symbols                          |         |         |           |
| Error compensation (optional) | PSK, QAM, ASK, APSK and MSK, measured signal                                        |         |         |         |         |          | estimated I/Q offset                             |         |         |           |
|                               |                                                                                     |         |         |         |         |          | estimated I/Q imbalance                          |         |         |           |
|                               |                                                                                     |         |         |         |         |          | estimated amplitude droop                        |         |         |           |
|                               |                                                                                     |         |         |         |         |          | estimated symbol rate error                      |         |         |           |
|                               |                                                                                     |         |         |         |         |          | estimated I/Q skew <sup>32, 33</sup>             |         |         |           |
|                               |                                                                                     |         |         |         |         |          | estimated channel response <sup>32</sup>         |         |         |           |
|                               | FSK, measured signal                                                                |         |         |         |         |          | estimated carrier frequency drift                |         |         |           |
|                               | FSK, reference signal                                                               |         |         |         |         |          | estimated FSK deviation error                    |         |         |           |
| Estimation points per symbol  | samples per symbol used for fine synchronization and equalizer estimation           |         |         |         |         |          | 1, 2 or capture oversampling                     |         |         |           |
| Swap I/Q                      | captured signal                                                                     |         |         |         |         |          | on/off                                           |         |         |           |

<sup>31</sup> Can never be larger than the set capture length. Furthermore, the maximum result length applies to the minimum capture oversampling. If the capture oversampling is larger, the maximum result length is reduced proportionally, e.g. for capture oversampling = 8, it is 32 000 symbols for the FSW.

<sup>32</sup> Not available for shaped offset QPSK.

<sup>33</sup> Not available for MSK.

## Display formats versus time

The following display formats versus time are available.

For this display format, the number of displayed samples per symbol is fixed to the selected capture oversampling.

|                               |  |                                |
|-------------------------------|--|--------------------------------|
| Captured signal <sup>34</sup> |  | magnitude versus time          |
|                               |  | I/Q versus time                |
|                               |  | absolute frequency versus time |

For these display formats, the parameter “display points per symbol” (1, 2, 4, 8, 16 or 32) sets the number of displayed samples per symbol.

|                                |                                                                                                                          |                                                   |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------|
| Measured signal <sup>35</sup>  | filtered, carrier locked, symbol locked                                                                                  | absolute/relative magnitude versus time           |
|                                |                                                                                                                          | I/Q versus time                                   |
|                                |                                                                                                                          | wrapped/unwrapped phase versus time               |
|                                |                                                                                                                          | absolute/relative frequency versus time           |
| Reference signal               | ideal, calculated from detected symbols                                                                                  | absolute/relative magnitude versus time           |
|                                |                                                                                                                          | I/Q versus time                                   |
|                                |                                                                                                                          | wrapped/unwrapped phase versus time               |
|                                |                                                                                                                          | absolute/relative frequency versus time           |
| Error vector signal            | vector difference between measured signal and reference signal                                                           | EVM versus time<br>(EVM normalization selectable) |
|                                |                                                                                                                          | I/Q versus time                                   |
| Error signal <sup>34, 35</sup> | difference between the measured signal's magnitude/phase/frequency and the reference signal's magnitude/phase/ frequency | magnitude error versus time                       |
|                                |                                                                                                                          | phase error versus time                           |
|                                |                                                                                                                          | absolute and relative frequency error versus time |

For all the listed results, spectrum and statistics (probability density function (PDF), cumulative probability density function (CDF), 95th percentile) are also available.

<sup>34</sup> The spectrum of the I/Q captured signal and the spectrum of the I/Q error vector can additionally be displayed together in one screen.

<sup>35</sup> The spectrum of the I/Q measured signal and the spectrum of the I/Q error vector can additionally be displayed together in one screen.

## Additional display formats

For this display format, the number of displayed samples per symbol is fixed to the selected capture oversampling.

|            |                 |               |
|------------|-----------------|---------------|
| I/Q vector | captured signal | polar diagram |
|------------|-----------------|---------------|

For these display formats, only results at symbol times are displayed.

|                             |                                   |                                                                                        |
|-----------------------------|-----------------------------------|----------------------------------------------------------------------------------------|
| I/Q constellation           | measured signal, reference signal | polar diagram<br>I/Q samples                                                           |
| I/Q constellation (rotated) | measured signal, reference signal | polar diagram<br>I/Q samples<br>(only for rotated constellations, e.g. $3\pi/8$ -8PSK) |
| Frequency constellation     | measured signal, reference signal | absolute frequency                                                                     |

For these display formats, the parameter “display points per symbol” (1, 2, 4, 8, 16 or 32) sets the number of displayed samples per symbol.

|                       |                                                           |                                                              |
|-----------------------|-----------------------------------------------------------|--------------------------------------------------------------|
| I/Q vector            | measured signal, reference signal,<br>error vector signal | polar diagram,<br>display of trajectory between symbol times |
| Frequency vector      | measured signal, reference signal                         | absolute frequency                                           |
| Eye diagram           | measured signal, reference signal                         | I eye diagram<br>Q eye diagram                               |
| Eye diagram frequency | measured signal, reference signal                         | eye diagram of absolute frequency                            |

## Display formats regarding the equalizer/channel

|                    |  |                                             |
|--------------------|--|---------------------------------------------|
| Impulse response   |  | magnitude<br>phase<br>I/Q versus time       |
| Frequency response |  | magnitude<br>phase<br>group delay           |
| Channel            |  | frequency response magnitude<br>group delay |

## Display of modulation accuracy results

The tables show the scalar result values calculated for each measurement. Additionally, the following statistical measures (calculated over multiple measurements) are shown for each result value: mean, peak (worst value), standard deviation and 95th percentile.

Limits can only be set for the current, mean and peak value of EVM, magnitude error, phase error, carrier frequency error, waveform quality factor and I/Q offset.

The tables are modulation-specific.

### Table for MSK, PSK, QAM, ASK and APSK

For the following results, the parameter “estimation points per symbol” can be set by the user. It can be set to 1 (only symbol times contribute to the result), 2 (two samples per symbol contribute to the result) or capture oversampling (all samples contribute to the result).

|                         |                                 |                                          |
|-------------------------|---------------------------------|------------------------------------------|
| I/Q offset              |                                 | calculation range automatically selected |
| I/Q imbalance           | not for BPSK, ASK               |                                          |
| Gain imbalance          | not for BPSK, ASK               |                                          |
| Quadrature error        | not for BPSK, ASK               |                                          |
| Amplitude droop         |                                 |                                          |
| Carrier frequency error |                                 |                                          |
| Symbol rate error       |                                 |                                          |
| I/Q skew                | not for MSK, shaped offset QPSK |                                          |

For the following results, the parameter “display points per symbol” can be set by the user. It can be set to 1 (only symbol times contribute to the result), 2 (two samples per symbol contribute to the result) or capture oversampling (all samples contribute to the result). The estimated I/Q offset, I/Q imbalance, amplitude droop, symbol rate error, I/Q skew may be optionally compensated before calculating these values.

|                                      |                                           |                                                       |
|--------------------------------------|-------------------------------------------|-------------------------------------------------------|
| Error vector magnitude (EVM)         | RMS and peak value of corresponding trace | user-settable calculation range<br>(evaluation range) |
| Modulation error ratio (MER)         | RMS and peak value of corresponding trace |                                                       |
| Magnitude error                      | RMS and peak value of corresponding trace |                                                       |
| Phase error                          | RMS and peak value of corresponding trace |                                                       |
| Mean power                           |                                           |                                                       |
| Waveform quality factor $\rho$ (rho) |                                           |                                                       |

Remark: For offset QPSK, the error vector magnitude (EVM) and modulation error ratio (MER) can be influenced by the parameter “offset EVM”.

### Table for FSK

For the following results, the parameter “estimation points per symbol” can be set by the user (1, 2 or capture oversampling).

|                           |  |                                          |
|---------------------------|--|------------------------------------------|
| FSK deviation error       |  | calculation range automatically selected |
| FSK measurement deviation |  |                                          |
| Carrier frequency drift   |  |                                          |
| Carrier frequency error   |  |                                          |

For the following results, the parameter “display points per symbol” can be set by the user (1, 2 or capture oversampling). The estimated FSK deviation error and the estimated carrier frequency drift may be optionally compensated before calculating these values.

|                 |                                           |                                                       |
|-----------------|-------------------------------------------|-------------------------------------------------------|
| Frequency error | RMS and peak value of corresponding trace | user-settable calculation range<br>(evaluation range) |
| Magnitude error | RMS and peak value of corresponding trace |                                                       |
| Mean power      |                                           |                                                       |

## Bit error rate

The bit error rate measurement requires one of the below mentioned prerequisites:

- An XML file containing all valid transmit sequences is loaded <sup>36</sup>
- The DUT transmits a PRBS bit sequence <sup>37</sup> (only for R&S®FSW3-KM103, R&S®FSW-K70P, R&S®FSWP-K70P, R&S®FSWT-K70P, R&S®FSMR3-K70P, R&S®FSV3-K70P, R&S®FPL1-K70P or R&S®VSE-K70P)

|                |  |                    |
|----------------|--|--------------------|
| Bit error rate |  | current value      |
|                |  | best-case value    |
|                |  | worst-case value   |
|                |  | accumulative value |

## Detected symbols

|                |                                                                                         |                                                                |
|----------------|-----------------------------------------------------------------------------------------|----------------------------------------------------------------|
| Symbol formats |                                                                                         | binary                                                         |
|                |                                                                                         | octal                                                          |
|                |                                                                                         | decimal                                                        |
|                |                                                                                         | hexadecimal                                                    |
| Symbol marker  | only if a pattern is loaded                                                             | correctly detected pattern symbols are marked in green         |
|                | only if prerequisites for bit error rate measurement are fulfilled (see Bit error rate) | decision errors within pattern are highlighted in red          |
|                |                                                                                         | decision errors within the result range are highlighted in red |

<sup>36</sup> The length of the transmit sequences needs to coincide with the length of the result range. A tool to record all valid transmit sequences is provided with firmware (R&S®VSA sequence recording).

Rohde & Schwarz recommends using an external trigger or a synchronization pattern to align the result range for this measurement.

<sup>37</sup> Supported PRBS orders: 7, 9, 11, 15, 16, 20, 21, 23, 31.

## Measurement uncertainty (nominal)

Specifications apply under the following conditions: temperature range from +20 °C to +30 °C; signal level  $\geq -25$  dBm; properly adjusted reference level; external reference frequency applied; offset between R&S®VSE/instrument center frequency and input signal center frequency smaller than 5 % of symbol rate; no additional I/Q impairments; random data sequence. Capture oversampling is set to 4. For symbol rates < 1 kHz or frequencies > 5 GHz, accuracy may be limited by phase noise.

### Residual errors for QPSK

The modulation is QPSK, the transmit filter is RRC with rolloff factor 0.22, the measurement filter is RRC with rolloff factor 0.22, and EVM is normalized to mean expected reference power. The parameter “estimation points per symbol” is set to 1, as is the parameter “display points per symbol” for the result summary. The result length is 150 symbols and the number of averages is 10.

| EVM                                  | Symbol rate            | FSW/FSWP/<br>FSWT | FSMR3000 | FSVA3000 | FSV3000  | FPS     | FPL     | RTO <sup>38</sup> (VSE) |
|--------------------------------------|------------------------|-------------------|----------|----------|----------|---------|---------|-------------------------|
| Residual EVM RMS<br>(averaged value) | 1 GHz center frequency |                   |          |          |          |         |         |                         |
|                                      | 100 kHz                | < 0.3 %           | < 0.3 %  | < 0.4 %  | < 0.5 %  | < 0.5 % | < 0.6 % | < 0.5 %                 |
|                                      | 1 MHz                  | < 0.4 %           | < 0.4 %  | < 0.45 % | < 0.5 %  | < 0.5 % | < 1.1 % | < 0.5 %                 |
|                                      | 10 MHz                 | < 0.6 %           | < 0.6 %  | < 0.65 % | < 0.7 %  | < 0.7 % | < 0.9 % | < 1.0 %                 |
|                                      | 20 MHz                 | < 1.0 %           | < 1.0 %  | < 1.0 %  | < 1.05 % | < 1.2 % | < 1.5 % | < 2.0 %                 |

| Carrier frequency error                                                                                                                                                        | Symbol rate            | FSW/FSWP/<br>FSWT | FSMR3000 | FSVA3000 | FSV3000 | FPS    | FPL    | RTO <sup>38</sup> (VSE) |
|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|-------------------|----------|----------|---------|--------|--------|-------------------------|
| Carrier frequency error uncertainty<br>(2 $\sigma$ value)                                                                                                                      | 1 GHz center frequency |                   |          |          |         |        |        |                         |
|                                                                                                                                                                                | 100 kHz                | 0.5 Hz            | 0.5 Hz   | 0.6 Hz   | 0.7 Hz  | 0.7 Hz | 1 Hz   | 10 Hz                   |
|                                                                                                                                                                                | 1 MHz                  | 3 Hz              | 3 Hz     | 4 Hz     | 5 Hz    | 6 Hz   | 10 Hz  | 15 Hz                   |
| The FSW/FSWP/<br>FSWT/FSMR3000/<br>FSVA3000/FSV3000/<br>FPS/FPL/RTO frequency<br>uncertainty needs to be added<br>separately.<br>Refer to the corresponding<br>specifications. | 10 MHz                 | 50 Hz             | 50 Hz    | 80 Hz    | 100 Hz  | 150 Hz | 200 Hz | 400 Hz                  |
|                                                                                                                                                                                | 20 MHz                 | 150 Hz            | 150 Hz   | 250 Hz   | 280 Hz  | 300 Hz | 350 Hz | 600 Hz                  |

### Residual errors for FSK

The modulation is 2FSK, the transmit filter is RRC with rolloff factor 0.2, the measurement filter is RRC with rolloff factor 0.2, and the FSK reference deviation is a quarter of the symbol rate. The parameter “estimation points per symbol” is set to 4 (capture oversampling), as is the parameter “display points per symbol” for the result summary. The result length is 150 symbols and the number of averages is 10.

| Frequency error                                  | Symbol rate            | FSW/FSWP/<br>FSWT | FSMR3000 | FSVA3000 | FSV3000 | FPS     | FPL     | RTO <sup>38</sup><br>(VSE) |
|--------------------------------------------------|------------------------|-------------------|----------|----------|---------|---------|---------|----------------------------|
| Residual frequency error RMS<br>(averaged value) | 1 GHz center frequency |                   |          |          |         |         |         |                            |
|                                                  | 100 kHz                | < 0.5 %           | < 0.5 %  | < 0.5 %  | < 0.5 % | < 0.5 % | < 0.5 % | < 0.5 %                    |
|                                                  | 1 MHz                  | < 0.5 %           | < 0.5 %  | < 0.5 %  | < 0.5 % | < 0.5 % | < 0.8 % | < 0.5 %                    |
|                                                  | 10 MHz                 | < 0.6 %           | < 0.6 %  | < 0.65 % | < 0.7 % | < 0.7 % | < 0.9 % | < 1.0 %                    |
|                                                  | 20 MHz                 | < 1.0 %           | < 1.0 %  | < 1.1 %  | < 1.2 % | < 1.2 % | < 1.5 % | < 2.0 %                    |

<sup>38</sup> Values apply for the speed optimized mode as mentioned in the R&S®VSE specifications.

## Residual errors for predefined standards

Measurements are based on the corresponding predefined standards. The number of averages is 10.

| EVM                                                 |                                       | FSW/FSWP/<br>FSWT | FSMR3000 | FSVA3000 | FSV3000 | FPS     | FPL     | RTO <sup>38</sup> (VSE) |
|-----------------------------------------------------|---------------------------------------|-------------------|----------|----------|---------|---------|---------|-------------------------|
| Residual EVM RMS<br>(averaged value)                | 1 GHz center frequency                |                   |          |          |         |         |         |                         |
|                                                     | 3GPP WCDMA (CPICH)                    | < 1.0 %           | < 1.0 %  | < 1.0 %  | < 1.0 % | < 1.0 % | < 1.0 % | < 1.0 %                 |
|                                                     | GSM EDGE (3π/8-8PSK,<br>normal burst) | < 0.4 %           | < 0.4 %  | < 0.45 % | < 0.5 % | < 0.5 % | < 0.6 % | < 0.6 %                 |
|                                                     | GSM (normal burst)                    | < 0.4 %           | < 0.4 %  | < 0.5 %  | < 0.6 % | < 0.6 % | < 0.8 % | < 0.8 %                 |
| Frequency error                                     |                                       | FSW/FSWP/<br>FSWT | FSMR3000 | FSVA3000 | FSV3000 | FPS     | FPL     | RTO <sup>38</sup> (VSE) |
| Residual frequency error<br>RMS<br>(averaged value) | 1 GHz center frequency                |                   |          |          |         |         |         |                         |
|                                                     | Bluetooth® (DH1)                      | < 0.8 %           | < 0.8 %  | < 0.8 %  | < 0.8 % | < 0.8 % | < 1.5 % | < 0.8 %                 |

## Ordering information

| Designation                                                                  | Type           | Order No.    |
|------------------------------------------------------------------------------|----------------|--------------|
| <b>Vector signal analysis</b>                                                |                |              |
| FSWX                                                                         |                |              |
| Vector signal analysis                                                       | R&S®FSW3-KM101 | 1347.7846.02 |
| Multi-modulation analysis, R&S®FSW3-KM101 option required                    | R&S®FSW3-KM102 | 1347.7852.02 |
| BER PRBS measurements, R&S®FSW3-KM101 option required                        | R&S®FSW3-KM103 | 1347.7869.02 |
| Cross-correlation for vector signal analysis, R&S®FSW3-KM101 option required | R&S®FSW3-KM104 | 1347.7875.02 |
| FSW                                                                          |                |              |
| Vector signal analysis                                                       | R&S®FSW-K70    | 1313.1416.02 |
| Multi-modulation analysis, R&S®FSW-K70 option required                       | R&S®FSW-K70M   | 1338.4177.02 |
| BER PRBS measurements, R&S®FSW-K70 option required                           | R&S®FSW-K70P   | 1338.3893.02 |
| R&S®FSWP                                                                     |                |              |
| Vector signal analysis, R&S®FSWP-B1 option required                          | R&S®FSWP-K70   | 1325.4280.02 |
| Multi-modulation analysis, R&S®FSWP-K70 option required                      | R&S®FSWP-K70M  | 1350.6860.02 |
| BER PRBS measurements, R&S®FSWP-K70 option required                          | R&S®FSWP-K70P  | 1350.6876.02 |
| R&S®FSWT                                                                     |                |              |
| Vector signal analysis                                                       | R&S®FSWT-K70   | 1338.7560.02 |
| Multi-modulation analysis, R&S®FSWT-K70 option required                      | R&S®FSWT-K70M  | 1350.5970.02 |
| BER PRBS measurements, R&S®FSWT-K70 option required                          | R&S®FSWT-K70P  | 1350.5986.02 |
| R&S®FSMR3000                                                                 |                |              |
| Vector signal analysis, R&S®FSMR3-B1 option required                         | R&S®FSMR3-K70  | 1345.3150.02 |
| Multi-modulation analysis, R&S®FSMR3-K70 option required                     | R&S®FSMR3-K70M | 1345.3737.02 |
| BER PRBS measurements, R&S®FSMR3-K70 option required                         | R&S®FSMR3-K70P | 1345.3720.02 |
| R&S®FSVA3000/FSV3000                                                         |                |              |
| Vector signal analysis                                                       | R&S®FSV3-K70   | 1330.5074.02 |
| Multi-modulation analysis, R&S®FSV3-K70 option required                      | R&S®FSV3-K70M  | 1346.3376.02 |
| BER PRBS measurements, R&S®FSV3-K70 option required                          | R&S®FSV3-K70P  | 1346.3382.02 |
| R&S®FPS                                                                      |                |              |
| Vector signal analysis                                                       | R&S®FPS-K70    | 1321.4127.02 |
| FPL                                                                          |                |              |
| Vector signal analysis                                                       | R&S®FPL1-K70   | 1323.1748.02 |
| Multi-modulation analysis, R&S®FPL1-K70 option required                      | R&S®FPL1-K70M  | 1323.1625.02 |
| BER PRBS measurements, R&S®FPL1-K70 option required                          | R&S®FPL1-K70P  | 1323.1631.02 |
| R&S®VSE                                                                      |                |              |
| Vector signal analysis measurement software                                  | R&S®VSE-K70    | 1320.7522.06 |
| Multi-modulation analysis, R&S®VSE-K70 option required                       | R&S®VSE-K70M   | 1345.1211.06 |
| BER PRBS measurements, R&S®VSE-K70 option required                           | R&S®VSE-K70P   | 1345.1228.06 |
| <b>Vector signal explorer software</b>                                       |                |              |
| Basic software                                                               | R&S®VSE        | 1345.1011.06 |
| Enterprise edition                                                           | R&S®VSE        | 1345.1105.06 |
| R&S®VSE software maintenance                                                 | R&S®VSE-SWM    | 1320.7622.81 |

| Designation                                          | Type         | Order No.    |
|------------------------------------------------------|--------------|--------------|
| <b>Analyzers</b>                                     |              |              |
| <b>FSWX</b>                                          |              |              |
| Signal and spectrum analyzer, 2 Hz to 26.5 GHz, 1 RF | R&S®FSWX3026 | 1348.0000.21 |
| Signal and spectrum analyzer, 2 Hz to 26.5 GHz, 2 RF | R&S®FSWX3026 | 1348.0000.22 |
| Signal and spectrum analyzer, 2 Hz to 44 GHz, 1 RF   | R&S®FSWX3044 | 1348.0000.41 |
| Signal and spectrum analyzer, 2 Hz to 44 GHz, 2 RF   | R&S®FSWX3044 | 1348.0000.42 |
| <b>FSW</b>                                           |              |              |
| Signal and spectrum analyzer, 2 Hz to 8 GHz          | R&S®FSW8     | 1331.5003.08 |
| Signal and spectrum analyzer, 2 Hz to 13.6 GHz       | R&S®FSW13    | 1331.5003.13 |
| Signal and spectrum analyzer, 2 Hz to 26.5 GHz       | R&S®FSW26    | 1331.5003.26 |
| Signal and spectrum analyzer, 2 Hz to 43.5 GHz       | R&S®FSW43    | 1331.5003.43 |
| Signal and spectrum analyzer, 2 Hz to 50 GHz         | R&S®FSW50    | 1331.5003.50 |
| Signal and spectrum analyzer, 2 Hz to 67 GHz         | R&S®FSW67    | 1331.5003.67 |
| Signal and spectrum analyzer, 2 Hz to 85 GHz         | R&S®FSW85    | 1331.5003.85 |
| <b>R&amp;S®FSWP</b>                                  |              |              |
| Phase noise analyzer, 1 MHz to 8 GHz                 | R&S®FSWP8    | 1322.8003.09 |
| Phase noise analyzer, 1 MHz to 26.5 GHz              | R&S®FSWP26   | 1322.8003.27 |
| Phase noise analyzer, 1 MHz to 50 GHz                | R&S®FSWP50   | 1322.8003.51 |
| <b>R&amp;S®FSVA3000</b>                              |              |              |
| Signal and spectrum analyzer, 10 Hz to 4 GHz         | R&S®FSVA3004 | 1330.5000.05 |
| Signal and spectrum analyzer, 10 Hz to 7.5 GHz       | R&S®FSVA3007 | 1330.5000.08 |
| Signal and spectrum analyzer, 10 Hz to 13.6 GHz      | R&S®FSVA3013 | 1330.5000.14 |
| Signal and spectrum analyzer, 10 Hz to 30 GHz        | R&S®FSVA3030 | 1330.5000.31 |
| Signal and spectrum analyzer, 10 Hz to 44 GHz        | R&S®FSVA3044 | 1330.5000.44 |
| <b>R&amp;S®FSV3000</b>                               |              |              |
| Signal and spectrum analyzer, 10 Hz to 4 GHz         | R&S®FSV3004  | 1330.5000.04 |
| Signal and spectrum analyzer, 10 Hz to 7.5 GHz       | R&S®FSV3007  | 1330.5000.07 |
| Signal and spectrum analyzer, 10 Hz to 13.6 GHz      | R&S®FSV3013  | 1330.5000.13 |
| Signal and spectrum analyzer, 10 Hz to 30 GHz        | R&S®FSV3030  | 1330.5000.30 |
| Signal and spectrum analyzer, 10 Hz to 44 GHz        | R&S®FSV3044  | 1330.5000.43 |
| <b>R&amp;S®FPS</b>                                   |              |              |
| Signal and spectrum analyzer, 9 kHz to 4 GHz         | R&S®FPS4     | 1319.2008.04 |
| Signal and spectrum analyzer, 9 kHz to 7 GHz         | R&S®FPS7     | 1319.2008.07 |
| Signal and spectrum analyzer, 9 kHz to 13.6 GHz      | R&S®FPS13    | 1319.2008.13 |
| Signal and spectrum analyzer, 9 kHz to 30 GHz        | R&S®FPS30    | 1319.2008.30 |
| Signal and spectrum analyzer, 9 kHz to 40 GHz        | R&S®FPS40    | 1319.2008.40 |
| <b>FPL</b>                                           |              |              |
| Spectrum analyzer, 5 kHz to 3 GHz                    | R&S®FPL1003  | 1304.0004.03 |
| Spectrum analyzer, 5 kHz to 7.5 GHz                  | R&S®FPL1007  | 1304.0004.07 |
| Spectrum analyzer, 5 kHz to 14 GHz                   | R&S®FPL1014  | 1304.0004.14 |
| Spectrum analyzer, 5 kHz to 26.5 GHz                 | R&S®FPL1026  | 1304.0004.26 |

| Designation                                                     | Type         | Order No.    |
|-----------------------------------------------------------------|--------------|--------------|
| <b>Test receiver</b>                                            |              |              |
| Test receiver, 10 Hz to 26.5 GHz                                | R&S®FSWT26   | 1313.7008.26 |
| <b>Measuring receivers</b>                                      |              |              |
| Measuring receiver, 10 Hz to 8 GHz                              | R&S®FSMR3008 | 1345.4004.08 |
| Measuring receiver, 10 Hz to 26.5 GHz                           | R&S®FSMR3026 | 1345.4004.26 |
| Measuring receiver, 10 Hz to 50 GHz                             | R&S®FSMR3050 | 1345.4004.50 |
| <b>Oscilloscopes</b>                                            |              |              |
| R&S®RTO1000                                                     |              |              |
| Oscilloscope, 600 MHz, 10 Gsample/s, 20/40 Msample, 2 channels  | R&S®RTO1002  | 1316.1000.02 |
| Oscilloscope, 600 MHz, 10 Gsample/s, 20/80 Msample, 4 channels  | R&S®RTO1004  | 1316.1000.04 |
| Oscilloscope, 1 GHz, 10 Gsample/s, 20/40 Msample, 2 channels    | R&S®RTO1012  | 1316.1000.12 |
| Oscilloscope, 1 GHz, 10 Gsample/s, 20/80 Msample, 4 channels    | R&S®RTO1014  | 1316.1000.14 |
| Oscilloscope, 2 GHz, 10 Gsample/s, 20/40 Msample, 2 channels    | R&S®RTO1022  | 1316.1000.22 |
| Oscilloscope, 2 GHz, 10 Gsample/s, 20/80 Msample, 4 channels    | R&S®RTO1024  | 1316.1000.24 |
| Oscilloscope, 4 GHz, 20 Gsample/s, 20/80 Msample, 4 channels    | R&S®RTO1044  | 1316.1000.44 |
| R&S®RTO2000                                                     |              |              |
| Oscilloscope, 600 MHz, 10 Gsample/s, 50/100 Msample, 2 channels | R&S®RTO2002  | 1329.7002.02 |
| Oscilloscope, 600 MHz, 10 Gsample/s, 50/200 Msample, 4 channels | R&S®RTO2004  | 1329.7002.04 |
| Oscilloscope, 1 GHz, 10 Gsample/s, 50/100 Msample, 2 channels   | R&S®RTO2012  | 1329.7002.12 |
| Oscilloscope, 1 GHz, 10 Gsample/s, 50/200 Msample, 4 channels   | R&S®RTO2014  | 1329.7002.14 |
| Oscilloscope, 2 GHz, 10 Gsample/s, 50/100 Msample, 2 channels   | R&S®RTO2022  | 1329.7002.22 |
| Oscilloscope, 2 GHz, 10 Gsample/s, 50/200 Msample, 4 channels   | R&S®RTO2024  | 1329.7002.24 |
| Oscilloscope, 3 GHz, 10 Gsample/s, 50/100 Msample, 2 channels   | R&S®RTO2032  | 1329.7002.32 |
| Oscilloscope, 3 GHz, 10 Gsample/s, 50/200 Msample, 4 channels   | R&S®RTO2034  | 1329.7002.34 |
| Oscilloscope, 4 GHz, 20 Gsample/s, 50/200 Msample, 4 channels   | R&S®RTO2044  | 1329.7002.44 |
| Oscilloscope, 6 GHz, 20 Gsample/s, 50/200 Msample, 4 channels   | R&S®RTO2064  | 1329.7002.64 |
| <b>External frontends</b>                                       |              |              |
| External frontend, 24 GHz to 44 GHz                             | R&S®FE44S    | 1338.7001.02 |
| External frontend, 36 GHz to 50 GHz                             | R&S®FE50DTR  | 1347.4099.02 |
| External frontend, 70 GHz to 110 GHz                            | R&S®FE110SR  | 1348.4840.02 |
| External frontend, 110 GHz to 170 GHz                           | R&S®FE170SR  | 1347.9090.02 |

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